HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python** scripts for ETL processes and automated reporting, processing approximately **5,000 records** per batch.
- Designed and implemented **Java Spring Boot** REST APIs for inter-service communication, handling up to **300 requests** per second.
- Contributed to the production support of enterprise applications, performing incident response and SLA management using tools like **Splunk** and **Dynatrace**.
- Built **Python** data pipelines on **AWS Lambda** to ingest and transform unstructured data, supporting downstream AI model development.
- Integrated **Large Language Models (LLMs)** into a proof-of-concept application using **OpenAI API** and **FastAPI**, enabling natural language querying.
- Implemented **Retrieval-Augmented Generation (RAG)** architecture using **LangChain** and **vector databases** to enhance response accuracy for a chatbot prototype.
- Containerized AI services using **Docker** for consistent deployment across development and staging environments.

SKILLS

AI/ML Frameworks – TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, LlamaIndex

Core Languages – Python, Java, SQL

AI/LLM Concepts – Generative AI, LLMs, Prompt Engineering, RAG, Embeddings, Vector Databases, AI Agents, Intelligent Automation

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins

APIs & Databases – REST APIs, SQL, PostgreSQL, MySQL, Oracle

Development Tools – Jira, Confluence, Model Evaluation, AI Application Deployment

PROJECTS

AI-Powered Document Analysis Service

- Developed a **Python** application integrating **OpenAI's GPT-4** model to analyze and summarize large volumes of unstructured text documents.
- Implemented **prompt engineering** techniques to optimize query output and reduce response latency by approximately **20%**.
- Designed a **REST API** using **FastAPI** to serve document analysis results, enabling integration with enterprise business systems.

Intelligent Chatbot with RAG Integration

- Built a conversational AI agent using **LangChain** and **Python** to answer user queries based on a curated knowledge base.
- Integrated a **vector database** (e.g., FAISS) and **embedding models** to enable semantic search and efficient information retrieval.
- Fine-tuned the retrieval mechanism, improving the accuracy of contextually relevant information by **15%** in user testing.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)